TITLE: LEDs interfacing with 8051 microcontroller for Different Operations.

AIM: To write simple programs in C

i) To understand interfacing of LEDs with 8051.

ii) To understand the use of Port of 8051 while programming.

THEORY:

LED: It is an electronic output device to display the status or logic levels of signal or data. LED's are available in different shapes, sizes and colors such as Red, Green, White, Yellow and Infrared as well.

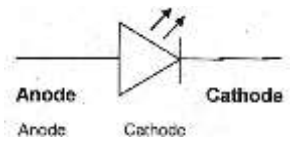


Figure 1: Structure of LED

ALGORITHM: -

Step 1: Start the program.
Step 2: Initialize Port 1 (P1) as an input port by writing 0xFF to it.
Step 3: Initialize Port 2 (P2) as an output port by clearing it or setting its initial state to 0x00.
Step 4: Read the current digital status of the switches connected to Port 1 into the accumulator or a register.

INTERFACING DIAGRAM:

CONCLUSION:

QUESTIONS:

1. Features of 8051 Microcontroller.
2. Draw and Explain Pin Diagram of 8051 Microcontroller.
3. What is IDE?
4. Write a C Program for Interfacing of 8 LED's to turn on or off by switches with 8051 Microcontroller.